

1

COMP

Water and solution properties

You should be able to: Explain how the polarity and hydrogen bonding of water determine solubility and identify whether a given solute is hydrophilic or hydrophobic.

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2**pH and buffers**

COMP

You should be able to: Define pH, state the normal pH range of arterial blood, and explain how a buffer pair resists a change in pH when acid or base is added.

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3

COMP

Protein structure and function

You should be able to: Relate the levels of protein structure to binding site shape and explain how denaturation by heat or pH change destroys function.

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4

COMP

Enzyme activity and regulation

You should be able to: Describe how enzymes lower activation energy and predict the effect of substrate concentration, temperature, pH, and competitive or allosteric inhibition on reaction rate.

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5

COMP

ATP and energy coupling

You should be able to: Explain how ATP hydrolysis is coupled to endergonic cellular work and name three categories of work that require it.

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6

COMP

Enzyme assay

You should be able to: Measure enzyme activity across a range of temperature or pH using a spectrophotometric or colorimetric assay, plot the results, and identify the optimum.

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7

COMP

ATP production pathways

You should be able to: Compare glycolysis, the citric acid cycle, and oxidative phosphorylation by location, oxygen requirement, and ATP yield.

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